KNECK OFFICIAL

Agriculture — Paper 3

443/3 • ORIGINAL REVISION SET 01 • Suggested 8 weeks (revision project) • 100 marks

Name: _____ Class: _____ Date: _____

Independent KCSE-style practice. Not issued, endorsed or predicted by KNEC. These are original questions, not leaked or past official papers. Component numbering is checked against the KNEC 2026 timetable. A teacher should review these practice materials before formal classroom assessment.

CANDIDATE INSTRUCTIONS

PROJECT component. This original shorter revision project is NOT the confidential official Agriculture project. The 2026 timetable provides a January–July project period. Conduct with school approval, record real observations honestly and label simulated data. Use non-hazardous practices.

Use a separate answer booklet. Show working where appropriate. The marking guidance begins after the END OF QUESTIONS page; do not consult it until you finish. Practical and project components require the stated equipment and teacher supervision, not merely a printed worksheet.

PROJECT — MULCH AND SOIL MOISTURE IN LEAFY VEGETABLES

Complete all phases; no prescribed result is assumed.

1. Plan an investigation of mulch versus no mulch on growth of a locally suitable leafy vegetable. State problem, hypothesis, objectives, risk assessment and work schedule. Choose one species and source healthy seedlings. [15 marks]

2. Design six equal plots or containers, three per treatment. Randomise treatment allocation; keep soil amount, seedling age, spacing, light, initial water and subsequent watering protocol consistent. Draw a labelled layout and explain the need for replication and controls. [20 marks]

3. Establish the crop. Apply a measured safe organic mulch layer to treatment plots without touching stems. Maintain all plots equally except mulch. Keep dated logs and photographs/drawings of key operations; do not fabricate evidence. [20 marks]

4. Collect weekly plant height and leaf count from at least three tagged plants per plot for six weeks. Record soil moisture using a consistent described method at fixed time relative to watering; rainfall/watering events and problems must be recorded. Include raw data tables, units and missing-data notes. [20 marks]

5. Calculate treatment means, draw labelled graphs and compare variation. Explain whether evidence supports the hypothesis and state at least three limitations; do not claim causal certainty from uncontrolled conditions. [15 marks]

6. Prepare a report including practical recommendations, a simple cost record and reflection. Present the work to a teacher and answer questions about method and data. [10 marks]

END OF QUESTIONS

MARKING GUIDANCE

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Suggested answers and assessment guidance, not an official marking scheme. Accept equivalent correct reasoning, valid examples and properly justified alternative methods. Marks for an item must not exceed its printed allocation. For essays use relevant textual evidence and the rubric; no plot summary alone.

PROJECT — MULCH AND SOIL MOISTURE IN LEAFY VEGETABLES

1. (15 marks) Clear testable question3; hypothesis/objectives4; feasible timing/resources3; safety/water/permission3; suitable species/seedlings2.

2. (20 marks) Layout/size4; random assignment4; three replicates4; controlled variables5; explanation bias/variability3.

3. (20 marks) Uniform establishment5; measured treatment5; consistent maintenance5; dated authentic evidence5. Do not reward favourable results over careful method.

4. (20 marks) Raw tables/units5; consistent repeated sampling6; moisture method/timing4; event/quality notes3; honest handling missing values2.

5. (15 marks) Correct means4; graphs4; evidence-linked conclusion3; limitations/variation4. Supplied or simulated data must never be represented as observed results.

6. (10 marks) Report/recommendations3; costs2; reflection2; authentic demonstration/questioning3.

REVISION CHECK

Mark your work, list three misconceptions, revise those topics and attempt the incorrect questions again without the guide. These topics are revision priorities, not a forecast of the real examination.

Paper/component reference: https://www.knec.ac.ke/wp-content/uploads/2026/02/2026-KCSE-Timetable.pdf